American Option

1 What Makes an Option "American"?

An **American option** grants the holder the right to exercise at any time up to and including the expiry date T. This is in contrast to a **European option**, which can only be exercised at expiry.

The flexibility to exercise early means an American option is always worth at least as much as the equivalent European option. The difference is called the **early exercise premium**:

$$V_{\text{American}} \geq V_{\text{European}}$$

2 The Intrinsic Value Floor

At any time t, the holder of an American option can exercise immediately and receive the intrinsic value. Because the option can always be surrendered for the intrinsic value, the price must satisfy:

$$\begin{aligned} V(S,t) &\geq \max(S-K,\ 0) \quad \text{(call)} \\ V(S,t) &\geq \max(K-S,\ 0) \quad \text{(put)} \end{aligned}$$

This floor is enforced at every node of the binomial tree, at every time step of the finite differences grid, and as a final floor in the Monte Carlo pricer.

3 When Is Early Exercise Optimal?

3.1 American Puts

For a put, early exercise becomes attractive when the option is sufficiently deep in-the-money. The intuition: if you hold a put with $K = 100$ on a stock currently at $S = 50$, you are owed \$50 in intrinsic value. Waiting means:

- **Benefit**: the stock might fall further, increasing the payoff
- **Cost**: you forgo the interest you could earn by investing the \$50 proceeds now

When the interest rate r is high and the option is deep enough in-the-money, the cost of waiting exceeds the benefit. Early exercise becomes optimal at the **early exercise boundary** $S^*(t)$ — the stock price below which the put should be exercised immediately.

3.2 American Calls — No Dividends

For a call on a **non-dividend-paying stock**, it is never optimal to exercise early. The argument is straightforward: holding the call is always at least as valuable as exercising it, because:

1. Exercising forfeits the remaining optionality ("insurance" against the stock falling)
2. The time value of delaying the strike payment K is positive

This is a classical result: an American call on a non-dividend stock has the same price as the corresponding European call.

$$V_{\text{Am. call}} = V_{\text{Eur. call}} \quad \text{when } q = 0$$

3.3 American Calls — With Dividends

When the stock pays a **continuous dividend yield** $q > 0$, holding the call means missing out on dividend income. For a sufficiently large dividend yield and deep in-the-money call, it can be optimal to exercise just before a large dividend payment. The early exercise premium for a call therefore depends on q.

4 Put-Call Parity Breaks Down

For European options, put-call parity gives an exact relationship:

$$C_{\text{Eur}} - P_{\text{Eur}} = Se^{-qT} - Ke^{-rT}$$

For American options, only an inequality holds:

$$Se^{-qT} - K \leq C_{\text{Am}} - P_{\text{Am}} \leq S - Ke^{-rT}$$

This means you cannot back out an American option price analytically from its counterpart via parity. Each side (call and put) must be priced independently.

5 No Analytical Solution

There is no closed-form formula for American option prices under the Black-Scholes model (except for the special case of a perpetual American put). Pricing requires numerical methods:

Method	Description
Binomial tree	Backward induction with early exercise check at each node
Finite differences	Solve the BS PDE with a free boundary (LCP)
Monte Carlo (LSM)	Longstaff-Schwartz regression-based optimal stopping

6 The `AmericanOption` Class

`AmericanOption` has the same interface as `EuropeanOption` — the distinction is purely about exercise style.

```
from options_pricer import AmericanOption, OptionType

# ATM 1-year put
put = AmericanOption(option_type=OptionType.PUT, strike=100.0, expiry=1.0)

# String coercion also works
call = AmericanOption(option_type="call", strike=100.0, expiry=0.5)
```

Validation mirrors `EuropeanOption`: both `strike` and `expiry` must be strictly positive.

```
from options_pricer import AmericanOption, BlackScholesModel, FiniteDifferencesPricer, MarketData, OptionTy

md    = MarketData(spot=100.0, rate=0.05)
model = BlackScholesModel(vol=0.20)

put = AmericanOption(option_type=OptionType.PUT, strike=100.0, expiry=1.0)

# Tree: ~5.89  (vs European ~5.57 - early exercise premium   0.32)
print(TreePricer(n_steps=500).price(put, md, model))

# Finite differences: ~5.89
print(FiniteDifferencesPricer().price(put, md, model))
```